

2.5.29

EE25BTECH11061 - V.Sainadh

Question: The points $\mathbf{A} = (4, 7)$, $\mathbf{B} = (p, 3)$, $\mathbf{C} = (7, 3)$ are the vertices of a right triangle, right-angled at $\mathbf{B}$. Find the value of p .

Solution: Let

$$\mathbf{A} = \begin{pmatrix} 4 \\ 7 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} p \\ 3 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 7 \\ 3 \end{pmatrix} \quad (1)$$

Right angle at $\mathbf{B} \implies$ the dot product of the adjacent sides at $\mathbf{B}$ is zero:

$$(\mathbf{A} - \mathbf{B})^T (\mathbf{C} - \mathbf{B}) = 0 \quad (2)$$

Compute the vectors:

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 4 - p \\ 4 \end{pmatrix}, \quad \mathbf{C} - \mathbf{B} = \begin{pmatrix} 7 - p \\ 0 \end{pmatrix} \quad (3)$$

Their dot product:

$$(\mathbf{A} - \mathbf{B})^T (\mathbf{C} - \mathbf{B}) = \begin{pmatrix} 4 - p \\ 4 \end{pmatrix}^T \begin{pmatrix} 7 - p \\ 0 \end{pmatrix} \quad (4)$$

$$= (4 - p)(7 - p) + 4 \cdot 0 \quad (5)$$

$$= (4 - p)(7 - p) \quad (6)$$

Right angle condition gives:

$$(4 - p)(7 - p) = 0 \implies p = 4 \text{ or } p = 7 \quad (7)$$

If $p = 7$, then $\mathbf{B} = \mathbf{C}$ (degenerate). Hence,

$$\boxed{p = 4} \quad (8)$$

Graph:

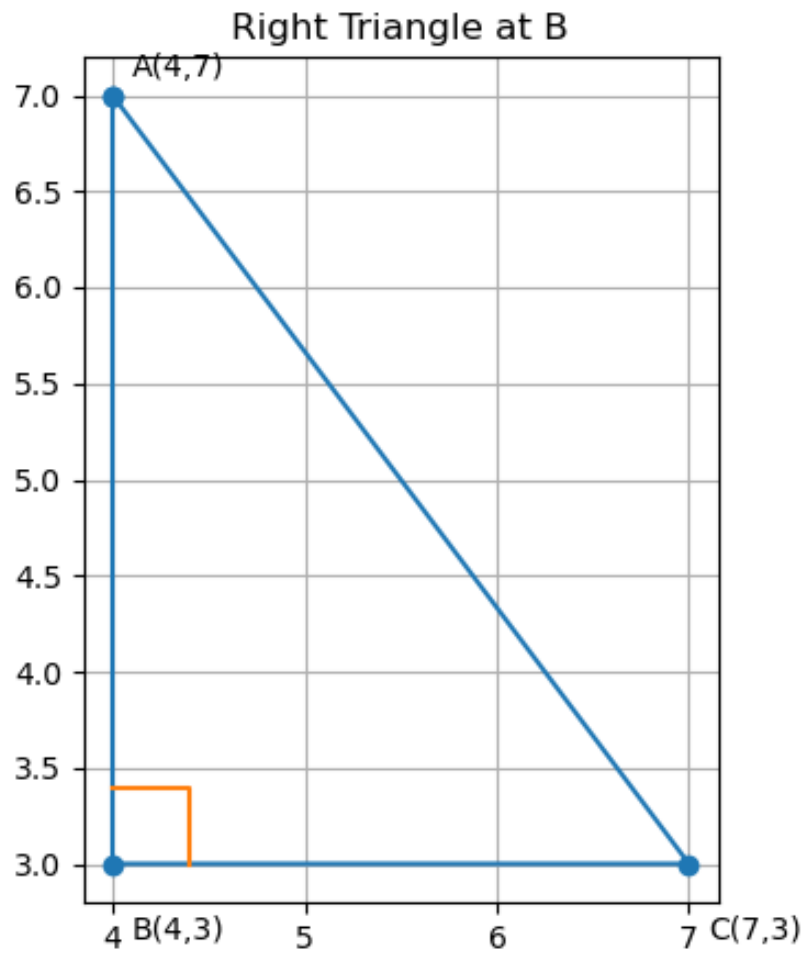


Fig. 1: Right Triangle at **B**